Chapter 9 Thermal Properties

To be Covered:

- How do materials respond to the application of heat?

- How do we define and measure...
  -- heat capacity?
  -- thermal expansion?
  -- thermal conductivity?
  -- thermal shock resistance?

- How do the thermal properties of ceramics, metals, and polymers differ?

- How to optimize the thermal conductivity in electrically insulating materials (ceramics) for microelectronic applications, e.g. SiC, Si_3N_4, diamond ?

- The relationship between the thermal properties of materials and their microstructure ?

Why we care about thermal properties of ceramics?

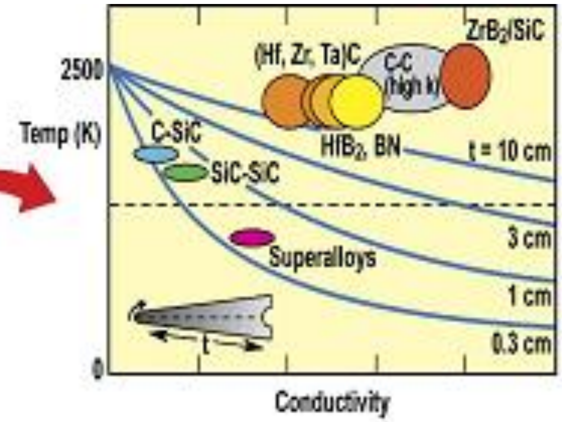
Sustained Hypersonic Flight Limited by Materials
- High heat flux over small area
- High temperature, oxidation, erosion
- Very high temperature gradients

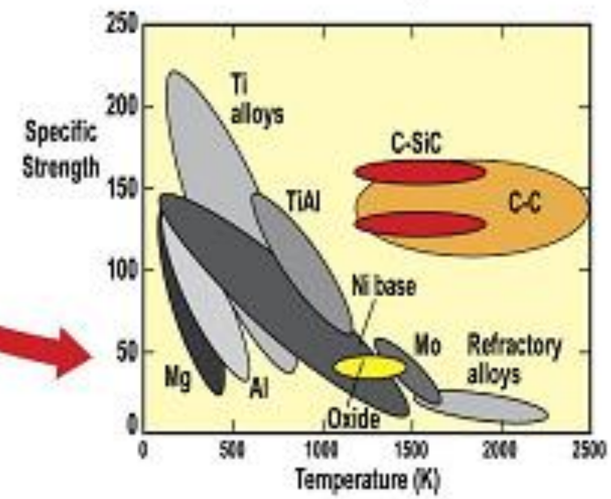
UHTCs(ZrB2/HfB2-based composites)

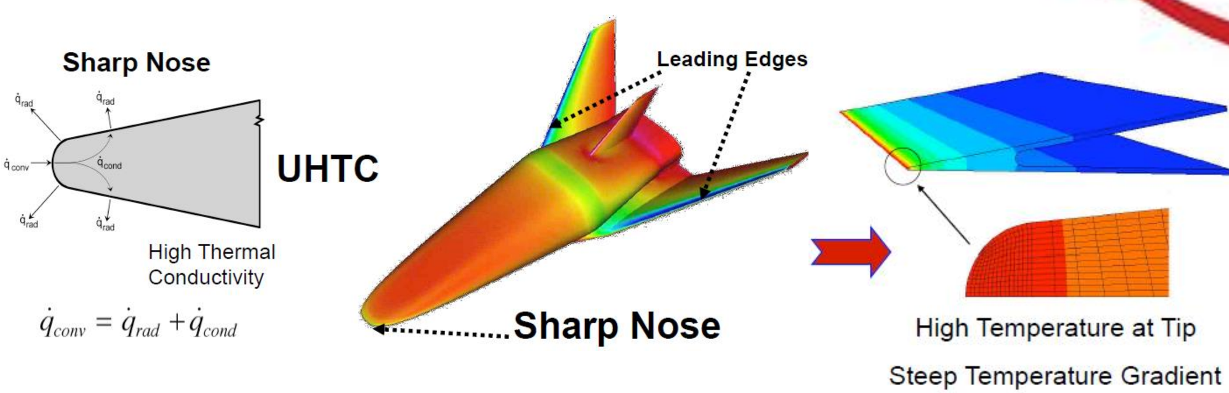
High Temperature capability

High Thermal conductivity

Aerospace Application

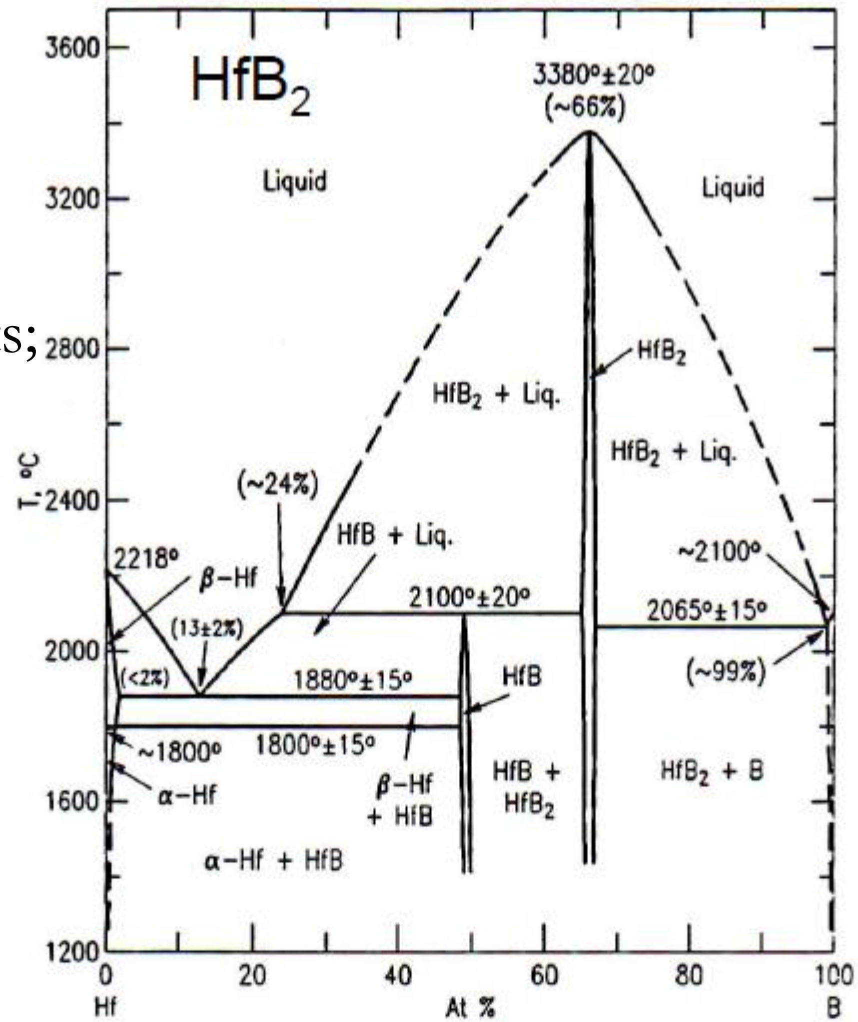






Ultra High Temperature Ceramics

- Borides, carbides and nitrides of transition elements such as Hafnium, Zirconium, Tantalum and Titanium;
- Some of highest known melting points;
- High hardness, good wear resistance, good mechanical strength;
- Good chemical and thermal stability under certain conditions;
- High thermal conductivity (dibordies) -good thermal shock resistance



Heat Capacity

The ability of a material to absorb heat

- Molar heat capacity: The energy required to produce a unit rise in temperature for one mole of a material.

Molar heat capacity (J/mol-K)

energy input (J/mol)

$$C = \frac{dQ}{dT}$$

temperature change (K)

- Specific heat: the amount of heat per unit mass required to raise the temperature by one degree Celsius.

$$C = \frac{dQ}{mdT}$$

- Two ways to measure heat capacity:

  C_p : Heat capacity at constant pressure.

  C_v : Heat capacity at constant volume.

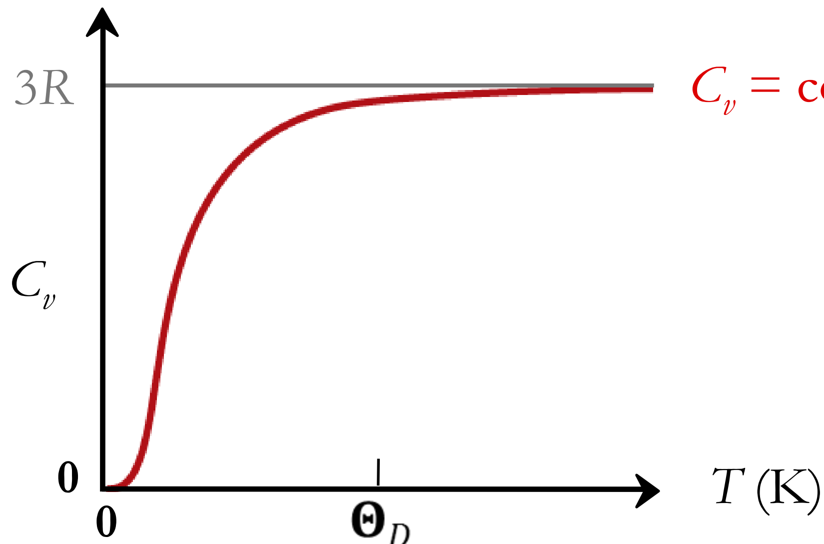
C_p usually > C_v

- Molar Heat capacity has units of $\dfrac{\mathrm{J}}{\mathrm{mol \cdot K}}$

Dependence of Heat Capacity on Temperature

- Heat capacity...
  -- increases with temperature
  -- for solids it reaches a limiting value of $3R$

R = gas constant
= 8.31 J/mol-K



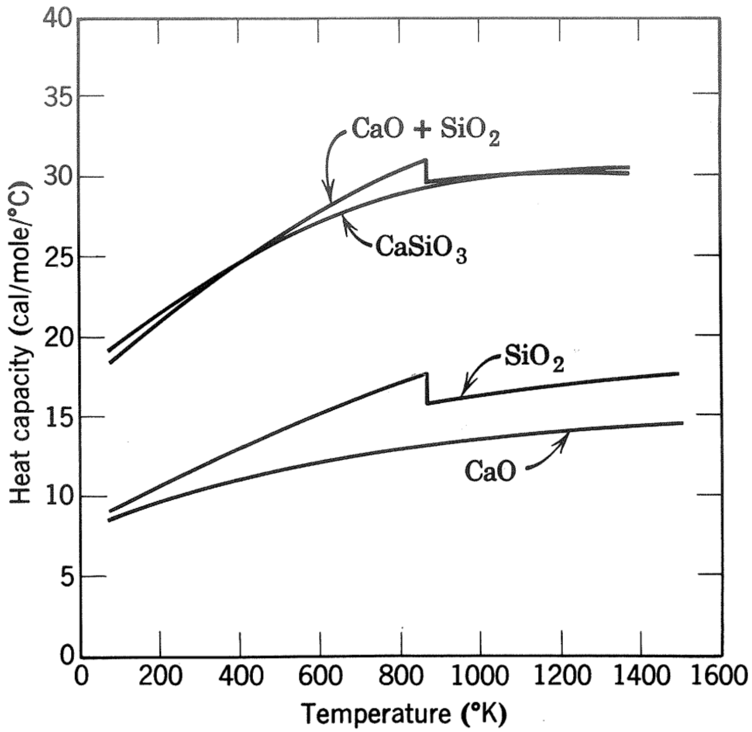
C_v = constant

Debye temperature
(usually less than T_{room})

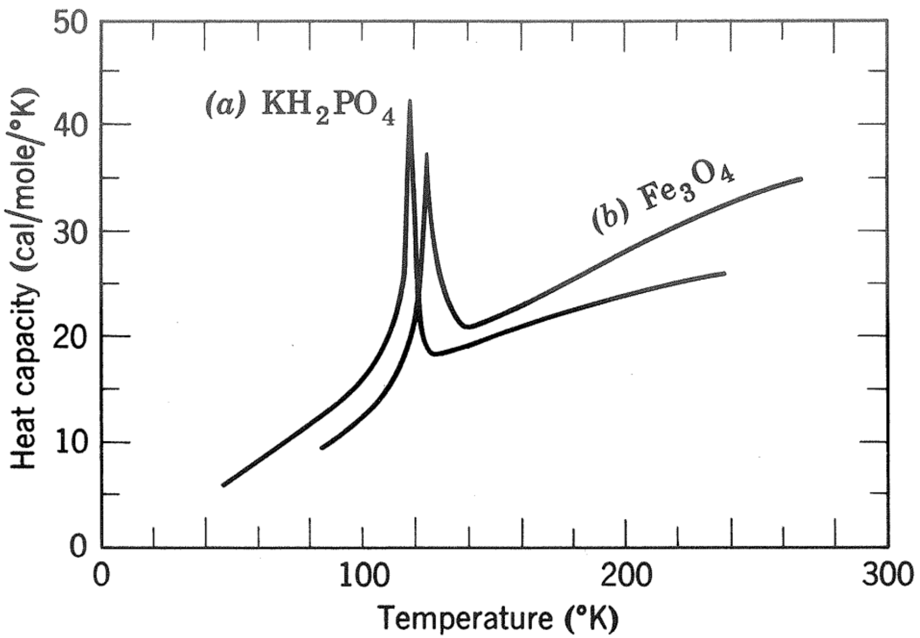
Adapted from Fig. 19.2, *Callister & Rethwisch 8e.*

- From atomic perspective:
  -- Energy is stored as atomic vibrations.
  -- As temperature increases, the average energy of atomic vibrations increases.

Crystal Structure and Ordered to Disordered Transformations



Heat capacity of various forms of CaO+SiO2 in 1:1 molar ratio



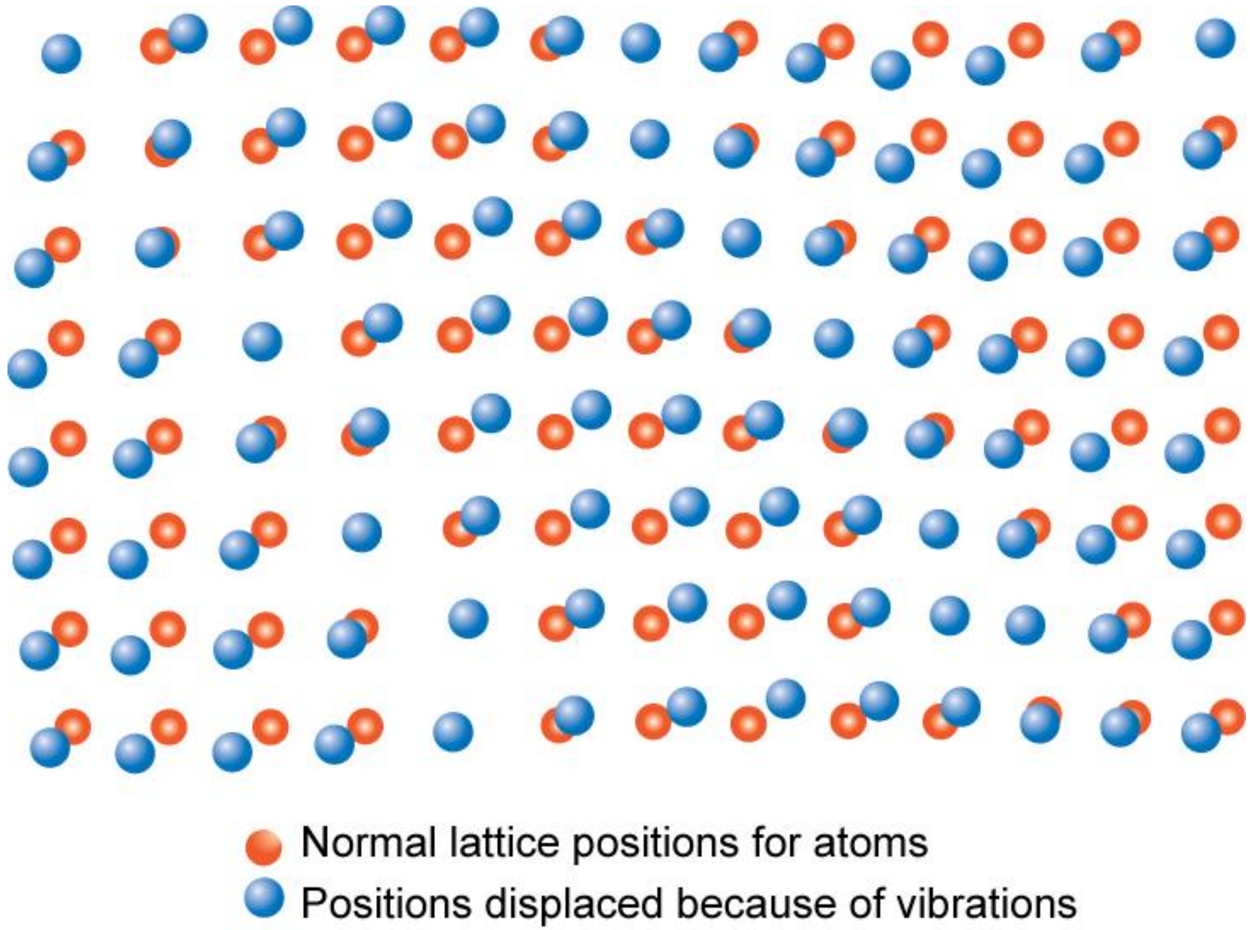
Heat capacity of order-disorder transformations of (a) hydrogen bonds in KH_2PO_4 and (b) Fe^{3+} ions in Fe_3O_4

Not structure-sensitive but depends on porosity

Increase rapidly during order-to-disorder transformation

Atomic Vibrations

Atomic vibrations are in the form of lattice waves or phonons



Adapted from Fig. 19.1,
Callister & Rethwisch 8e.

Specific Heat: Comparison

increasing c_p

Material	c_p (J/kg-K) at room T
• Polymers	
Polypropylene	1925
Polyethylene	1850
Polystyrene	1170
Teflon	1050
• Ceramics	
Magnesia (MgO)	940
Alumina (Al_2O_3)	775
Glass	840
• Metals	
Aluminum	900
Steel	486
Tungsten	138
Gold	128

c_p (specific heat): (J/kg-K)
C_p (heat capacity): (J/mol-K)

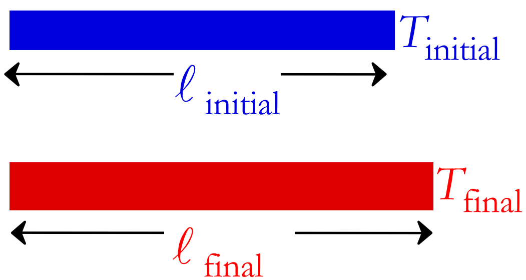
- Why is c_p significantly larger for polymers?

Selected values from Table 19.1, *Callister & Rethwisch 8e.*

Thermal Expansion

Materials change size when temperature is changed

Coefficient of linear expansion:



$$\alpha_\ell = \frac{d\ell}{\ell dT}$$

$$T_{final} > T_{initial}$$

$$\frac{\ell_{final} - \ell_{initial}}{\ell_{initial}} = \overline{\alpha_\ell}(T_{final} - T_{initial})$$

linear coefficient of
thermal expansion (1/K or 1/°C)

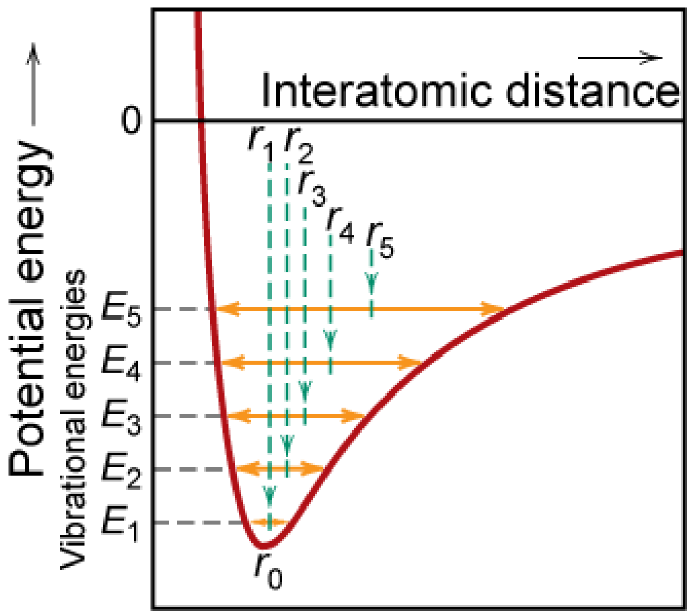
Coefficient of volume expansion:

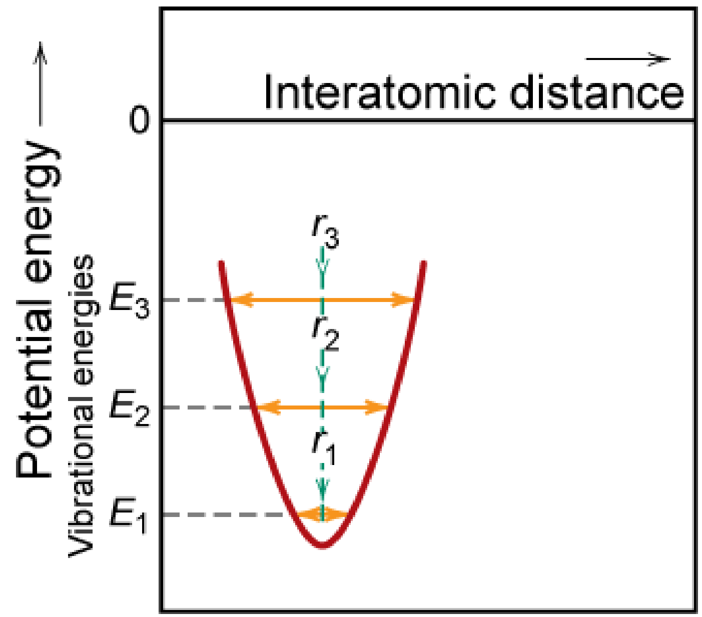
$$\alpha_V = \frac{dV}{VdT}$$

$$\frac{V_{final} - V_{initial}}{V_{initial}} = \overline{\alpha_V}(T_{final} - T_{initial})$$

http://global.kyocera.com/fcworld/charact/heat/thermaexpan.html

Atomic Perspective: Thermal Expansion





Asymmetric curve:
- -- increase temperature,
- -- increase in interatomic separation
- -- thermal expansion

Symmetric curve:
- -- increase temperature,
- -- no increase in interatomic separation
- -- no thermal expansion

Adapted from Fig. 19.3, *Callister & Rethwisch 8e.*

Coefficient of Thermal Expansion: Comparison

Material	a_ℓ (10^{-6}/°C) at room T
• Polymers	
Polypropylene	145-180
Polyethylene	106-198
Polystyrene	90-150
Teflon	126-216
• Metals	
Aluminum	23.6
Steel	12
Tungsten	4.5
Gold	14.2
• Ceramics	
Magnesia (MgO)	13.5
Alumina (Al_2O_3)	7.6
Soda-lime glass	9
Silica (cryst. SiO_2)	0.4
Si3N4	2.6
SiC	4.4

Polymers have larger α_ℓ values because of weak secondary bonds

- **Q: Why does a_ℓ generally decrease with increasing bond energy?**

Selected values from Table 19.1, *Callister & Rethwisch 8e.*

Thermal Expansion: Example

Ex: A copper wire 15 m long is cooled from 40 to -9°C. How much change in length will it experience?

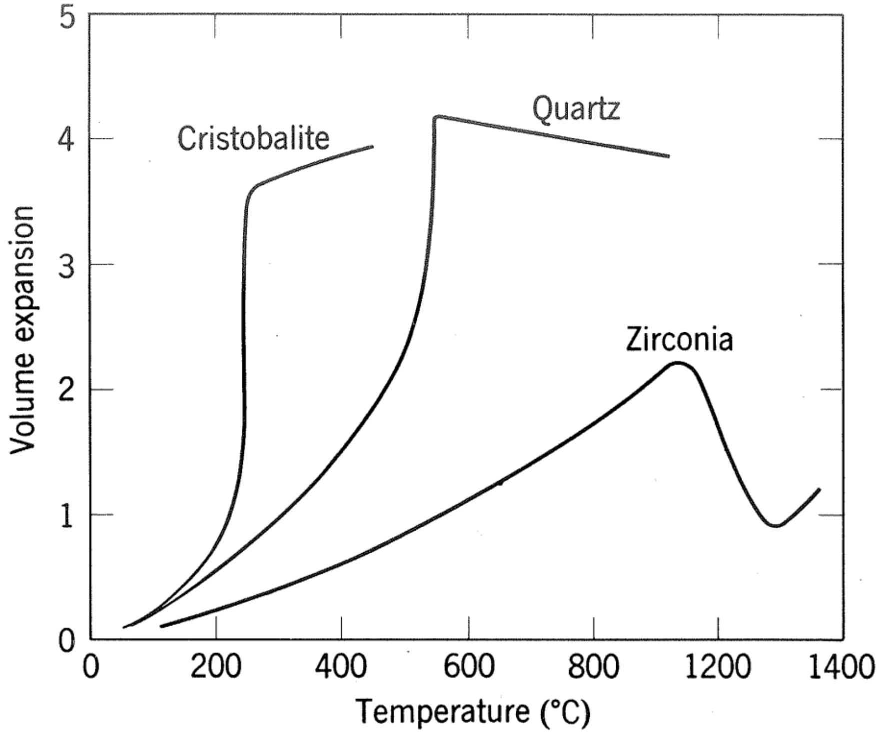
- Answer: For Cu

$$\alpha_\ell = 16.5 \text{ x } 10^{-6} \; (^\circ\text{C})^{-1}$$

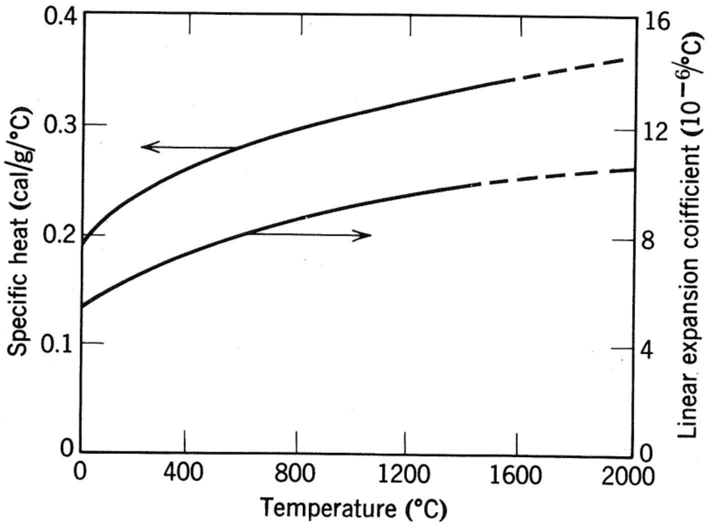
$$\Delta\ell = \alpha_\ell \ell_0 \Delta T = [16.5 \text{ x } 10^{-6} (1/^\circ\text{C})](15\text{ m})[40^\circ\text{C} - (-9^\circ\text{C})]$$

$$\boxed{\Delta\ell = 0.012\text{ m} = 12\text{ mm}}$$

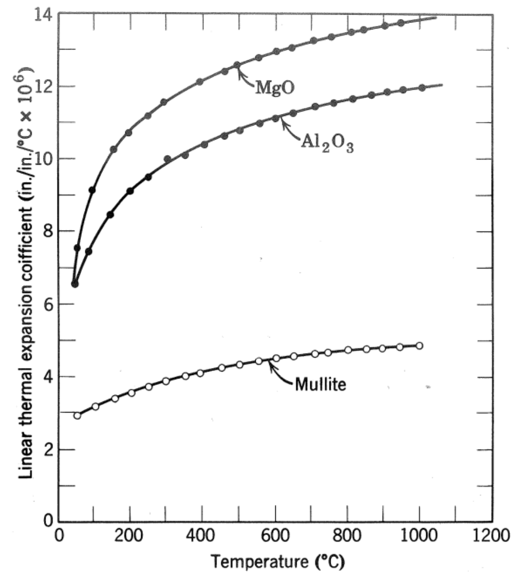
Density and thermal expansion of crystals



Thermal expansion coefficient versus temperature



Parallel changes in heat capacity and thermal expansion coefficient of Al_2O_3



Thermal expansion coefficient versus temperature for some ceramic oxides.

For isotropic material, the average volume expansion coefficient is related to the linear expansion coefficient by

$$1 + \bar{\alpha}\,\Delta T = 1 + 3\bar{a}\,\Delta T + 3\bar{a}^2\,\Delta T^2 + \bar{a}^3\,\Delta T^3$$

$$\bar{\alpha} = 3\bar{a} + 3\bar{a}^2\,\Delta T + \bar{a}^3\,\Delta T^2$$

$$\boxed{\bar{\alpha} = 3\bar{a}}$$

For anisotropic Materials

Crystal	Normal to c-Axis	Parallel to c-Axis
Al_2O_3	8.3	9.0
Al_2TiO_5	−2.6	+11.5
$3Al_2O_3 \cdot 2SiO_2$	4.5	5.7
TiO_2	6.8	8.3
$ZrSiO_4$	3.7	6.2
$CaCO_3$	−6	25
SiO_2 (quartz)	14	9
$NaAlSi_3O_8$ (albite)	4	13
C (graphite)	1	27

Thermal Expansion of Composite Bodies

When a ploycrystalline body, a mixture of crystalline phases, or a mixture of crystals and glass:

(1) If thermal expansion coefficient in different crystalline directions are not the same;

(2) If the various phase have different coefficients of thermal expansion;

(3) If different grains present have different amounts of contraction on cooling;

Thermal Expansion of Composite Bodies

The stress on each particle

$$\sigma_i = K(\alpha_r - \alpha_i)\,\Delta T$$

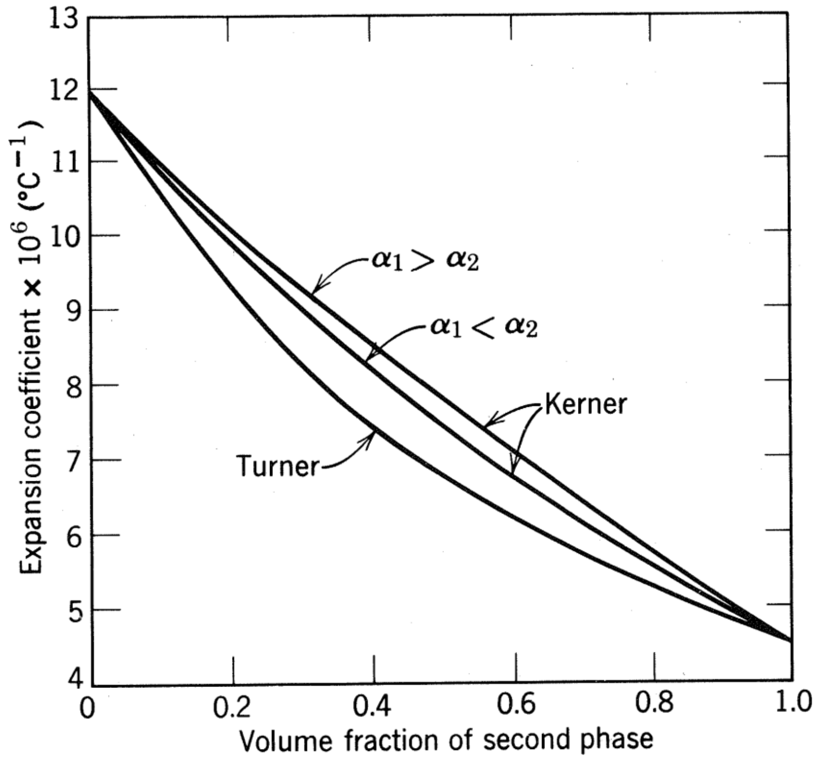
The summation of the stresses over the volume is zero

$$K_1(\alpha_r - \alpha_1)V_1\,\Delta T + K_2(\alpha_r - \alpha_2)V_2\,\Delta T + \cdots = 0$$

$$V_1 + V_2 + \cdots = V_r \qquad V_i = \frac{F_i \rho_r V_r}{\rho_i}$$

The coefficient of volume expansion of the aggregate by Turner

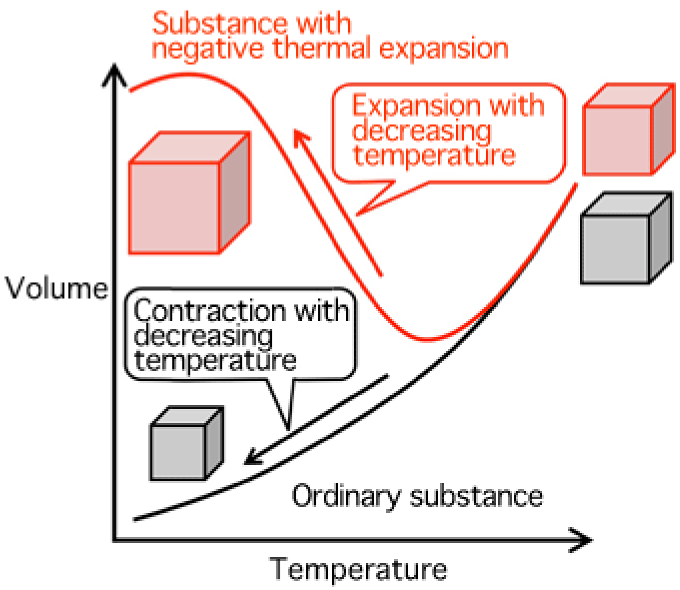
$$\alpha_r = \frac{\alpha_1 K_1 F_1/\rho_1 + \alpha_2 K_2 F_2/\rho_2 + \cdots}{K_1 F_1/\rho_1 + K_2 F_2/\rho_2 + \cdots}$$



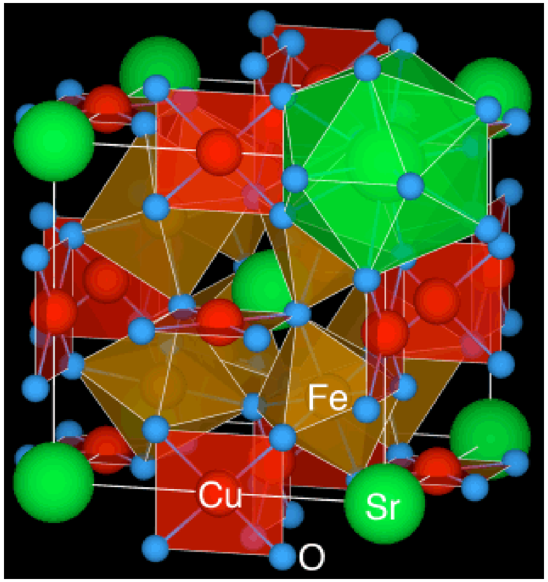
Taking into account shear effects at the boundaries between grains or phases, the overall expansion coefficient can be expressed by the relation first obtained by Kerner

$$\alpha_r = \alpha_1 + V_2(\alpha_2 - \alpha_1)\frac{K_1(3K_2 + 4G_1)^2 + (K_2 - K_1)(16G_1^{\,2} + 12G_1K_2)}{(4G_1 + 3K_2)[4V_2G_1(K_2 - K_1) + 3K_1K_2 + 4G_1K_1]}$$

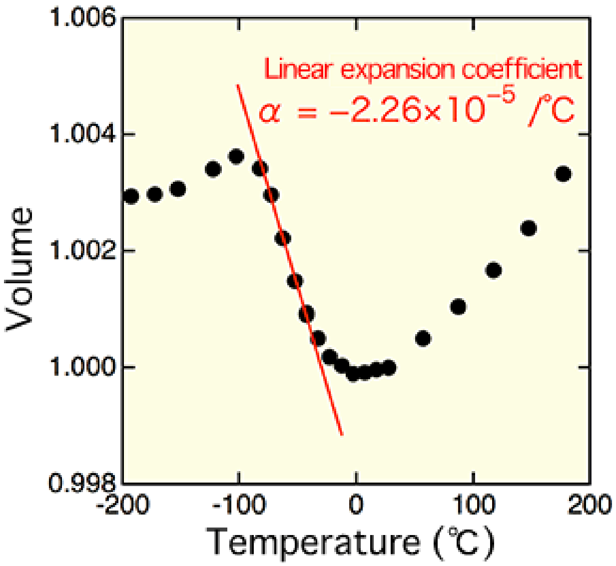
Negative / Zero Thermal Expansion Possible ?



Schematic of positive and negative thermal expansion



Crystal structure of $SrCu_3Fe_4O_{12}$



Change in volume of $SrCu_3Fe_4O_{12}$ with temperature

Negative thermal expansion

Some substances contract with increasing temperature and expand with decreasing temperature. Because the thermal expansion rate is negative for these substances, this phenomenon is called negative thermal expansion. This phenomenon is very important for the development of zero-thermal-expansion (see below) materials that do not contract or expand when the temperature changes.

Zero thermal expansion

no expansion or contraction with a change in temperature. Materials with zero thermal expansion may be manufactured by combining substances with positive thermal expansion and those with negative thermal expansion.

Thermal Conductivity

The ability of a material to transport heat.

Fourier's Law

$$q = -k\frac{dT}{dx}$$

heat flux
(J/m²-s)

temperature gradient

thermal conductivity (J/m-K-s)

$T_2 > T_1$

- Atomic perspective: Atomic vibrations and free electrons in hotter regions transport energy to cooler regions.

Thermal Conductivity- basics

• The kinetic theory of gases provides a basic model for which thermal conductivity, k, depends on the specific heat, C, the velocity, v, [of the carrier of heat] and the mean free path, l.

$$k = 1/3\, C\, v\, l$$

• In solids, the heat carrier is the phonon and so the relevant mean free path is the distance between phonon scattering centers. Collisions between phonons are not relevant because they do not change the net momentum of the phonons involved in transport whereas so-called umklapp processes, in which two phonons interact with the lattice to yield a third (scattered) phonon. These processes change the net momentum of the phonons, precisely because of the interaction with the lattice.

• Recall that sound velocities are related to elastic stiffness, E, and density,

$$v = \sqrt{E / \rho}$$

Thermal Conductivity- basics

• It turns out that at high temperatures, the number of phonons involved in such scattering is proportional to T, from which it follows that the mean free path goes as 1/T.

• Also, as the temperature decreases and the mean free path goes up, eventually the length is limited by either the specimen size or the grain size.

• At low enough temperatures, the specific heat varies strongly with temperature, $C \propto T^3$. Thus the thermal conductivity also varies at T^3.

High *k* Ceramics

- Phonons are elastic waves in the crystal lattice.

Heat is mostly carried by phonons in the acoustic range, i.e. with frequencies below 20kHz (or 1000 cm^{-1}).

- Srivastava gives an equation for thermal conductivity at high temperatures from three-phonon (scattering) interactions, where <M> is the average atomic mass, Ω is the atomic volume, B is a constant, ΘD is the Debye temperature (~516K for AlN), T is the temperature, and γ is the Grüneisen constant (~0.77 for AlN).

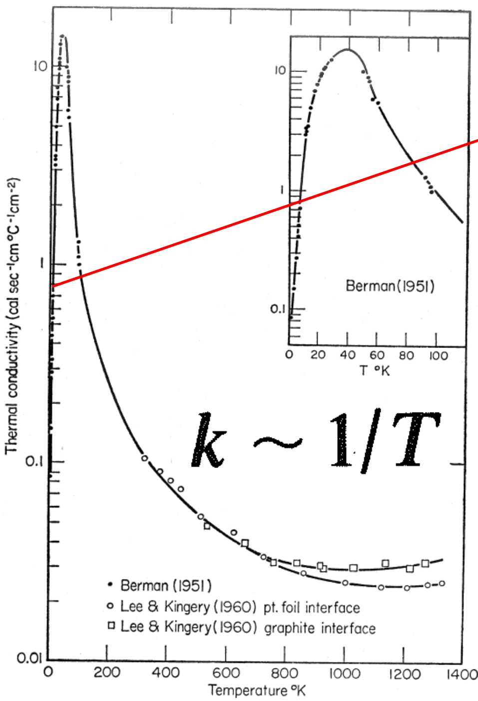
$$k = B\langle M\rangle\Omega^{1/3}\Theta^3/(T\gamma^2)$$

- Based on this equation, the following factors promote high thermal conductivity.

– Low atomic mass
– Strong interatomic bonding (covalent in preference to ionic) for high Debye temperature
– Equivalently, high modulus
– Simple crystal structure (less scattering of phonons)
– Low anharmonicity
– Isotopically pure

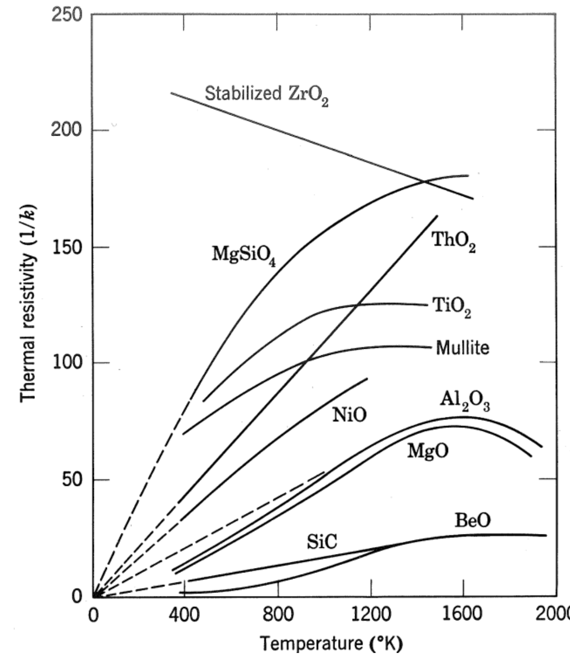
- Examples: Diamond 2000 ($W.m^{-1}.K^{-1}$) BN (cubic) 1300 SiC 490 BeO 370 AlN 320

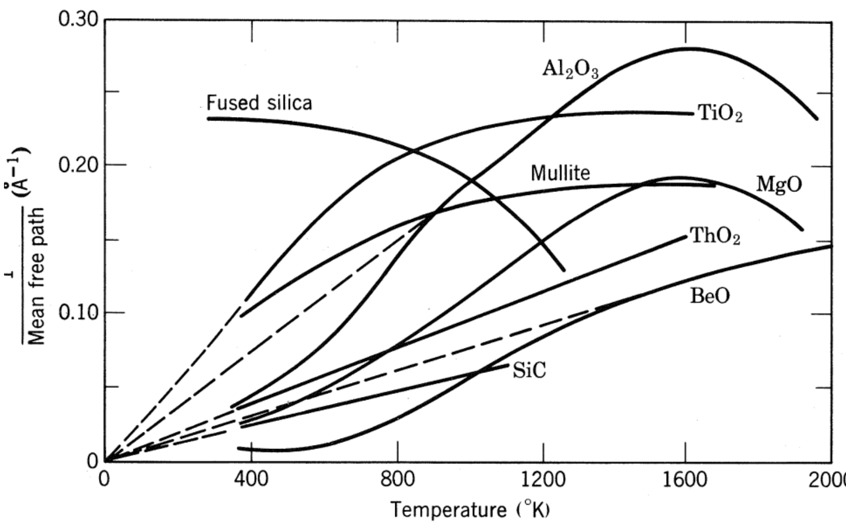
Temperature dependence



$$k \sim \exp(-\theta/\alpha T)$$

$$k \sim 1/T$$





Defects

Heat sinks are typically made by sintering powders together.

• The important defects are the following.

– Point defects in the lattice

– Grain boundaries

– Pores

– Second phases from dopants, sintering aids

• Grain size: just as in mechanical strength (Hall-Petch effect), the increase in thermal resistivity, k-1, is related to the the square root of the grain size,$\sqrt{d}$:

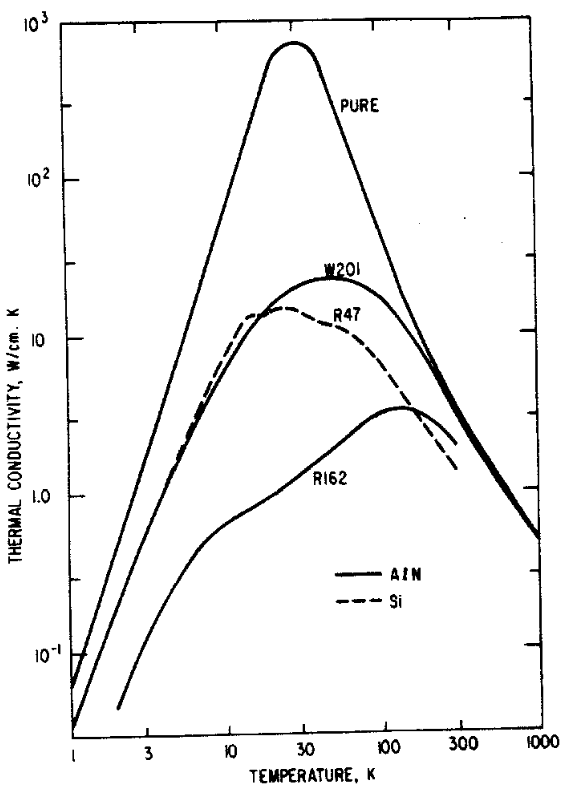
$$\Delta(k^{-1}) \propto \sqrt{d}$$

Point Defects

The importance of point defects depends on the material under consideration.

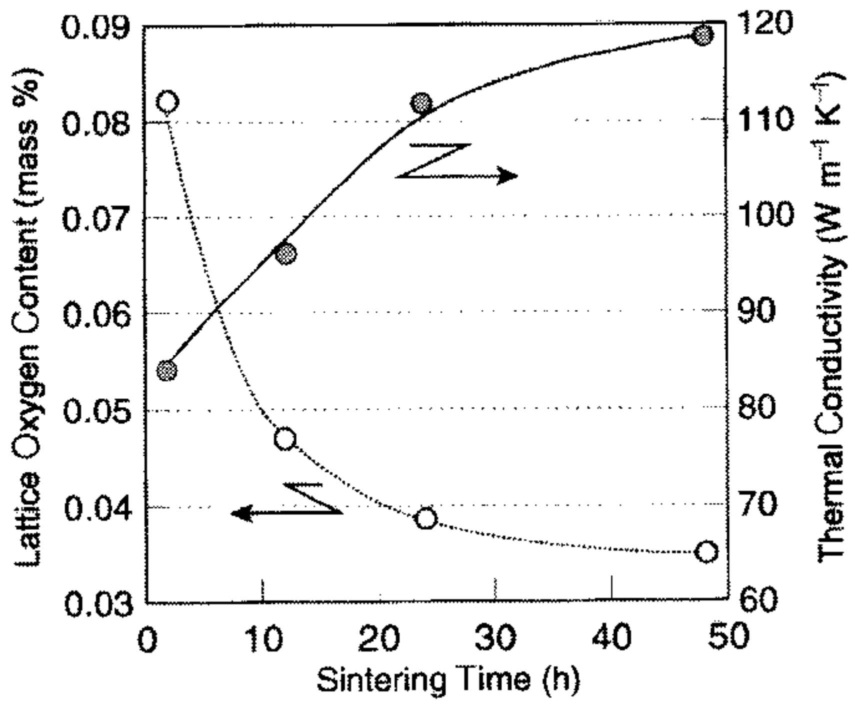
- For example, in silicon nitride, the oxygen content is critical.

Oxygen content is minimized by sintering under a nitrogen atmosphere.



Oxygen contents (in 10^{18} atoms.cm^{-3}):
W201: 42;
R47 (Si): 1;
R162: 300

Conductivity goes down with increasing oxygen content.

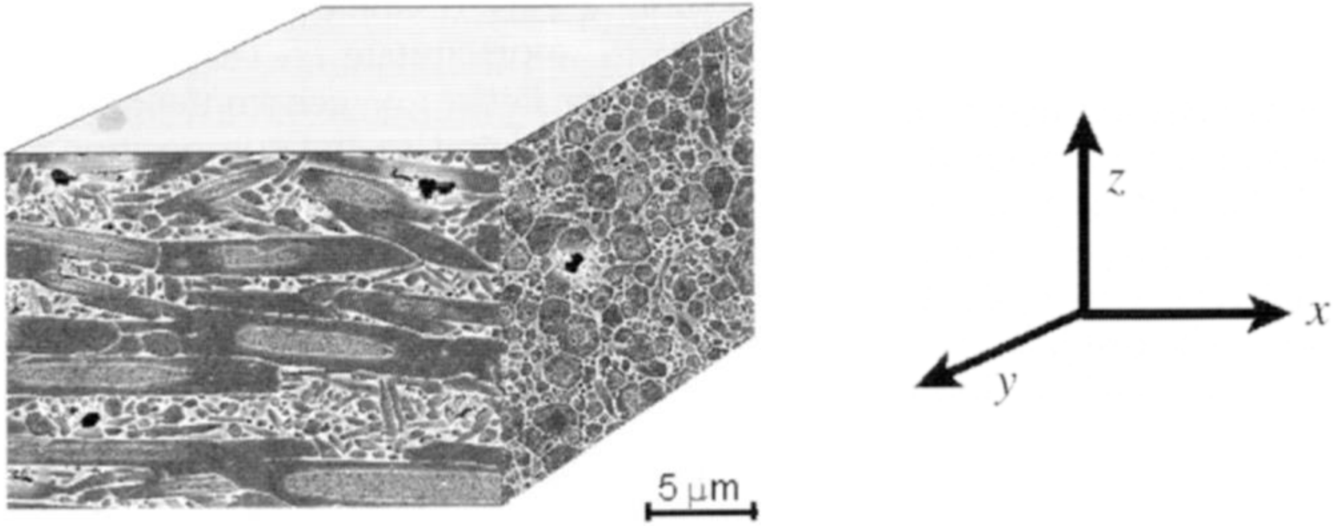


Effect of sintering time on the lattice oxygen content and the thermal conductivity of Si_3N_4 with 2 mol.% Yb_2O_3-5mol.% MgO sintered at 1900˚C under 0.9 Mpa N_2

Aligned Microstructures

- Which direction has the highest conductivity? Lowest?
- Si_3N_4 made by combined seeding (with β-Si_3N_4) and tape casting.

Sintering aids are Y_2O_3 or Nd_2O_3,nitrogen atmosphere.

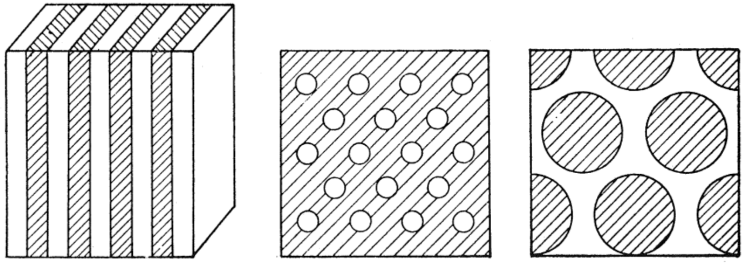


Microstructure of textured Si_3N_4 ceramic

Thermal Conductivity of Ceramics

- Most ionically and covalently bonded materials exhibit very low conductivities (electrical and thermal). This is a result of the large band gap between the valence and conduction bands.
- Covalently bonded solids tend to have higher thermal conductivities than ionically bonded solids.
- Insulating materials with high thermal conductivity are useful, however, for electronic applications where they are used to conduct heat away from components with high heat dissipation requirements.
- Adamantine materials (diamond-like) have thermal conductivities comparable to those of fcc metals.
- Thermal waves can be used to measure k, e.g. by irradiating a sample with a laser pulse and observing the change in temperature on the other side.

Conductivity of Multiphase Ceramics



(a) If the heat flow is parallel to slabs,

$$k_m = v_1 k_1 + v_2 k_2$$

If the heat flow is perpendicular l to slabs,

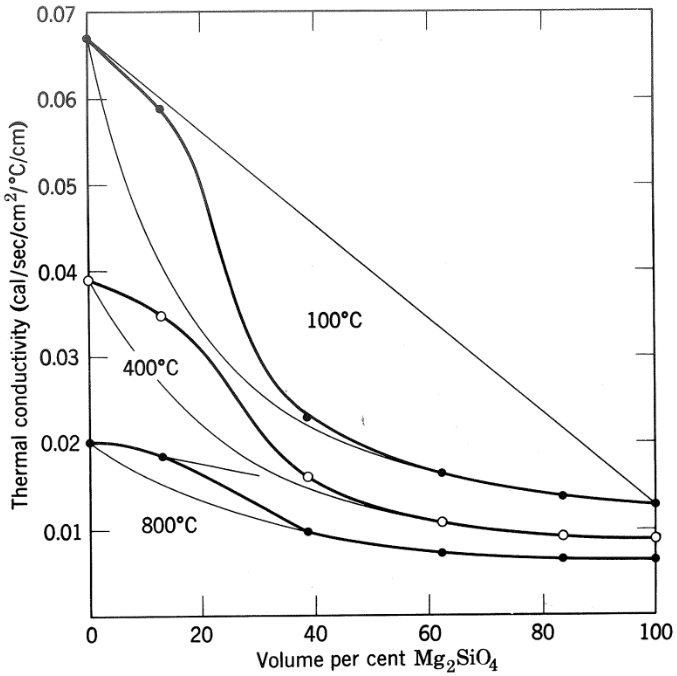
$$\frac{1}{k_m} = \frac{v_1}{k_1} + \frac{v_2}{k_2} \qquad k_m = \frac{k_1 k_2}{v_1 k_2 + v_2 k_1}$$

(b) & (c)

$$k_m = k_c \frac{1 + 2v_d(1 - k_c/k_d)/(2k_c/k_d + 1)}{1 - v_d(1 - k_c/k_d)/(k_c/k_d + 1)}$$

k_m and k_c are conductivity of continuous phase and dispersed phase respectively

When $k_c > k_d$, the resultant conductivity is $k_m \simeq k_c[(1 - v_d)/(1 + v_d)]$. In contrast, if $k_d > k_c$, then $k_m \simeq k_c[(1 + 2v_d)/(1 - v_d)]$.



Thermal conductivity in the two-phase system MgO-$MgSiO_4$

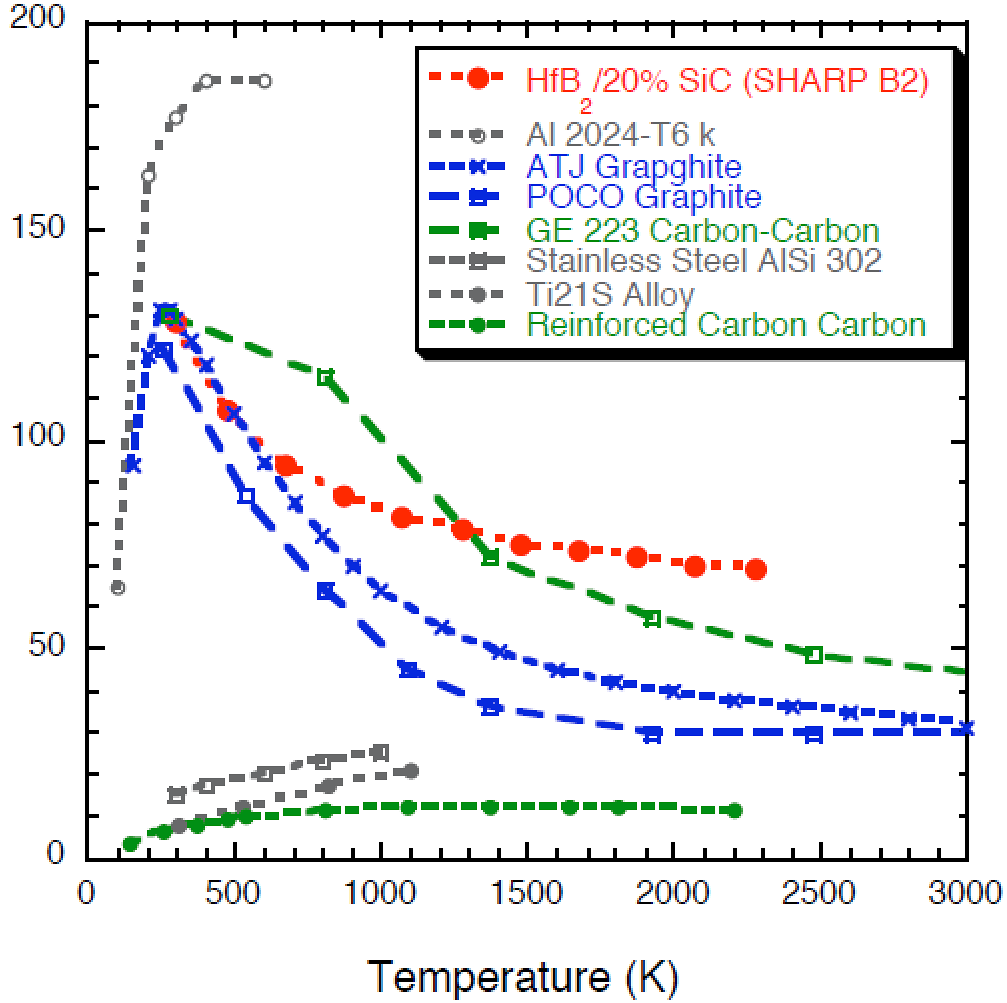
Thermal Conductivity Comparison

increasing k

Material	k (W/m-K)	Energy Transfer Mechanism
• Metals		
Aluminum	247	atomic vibrations and motion of free electrons
Steel	52	
Tungsten	178	
Gold	315	
• Ceramics		
Magnesia (MgO)	38	atomic vibrations
Alumina (Al_2O_3)	39	
Soda-lime glass	1.7	
Silica (cryst. SiO_2)	1.4	
• Polymers		
Polypropylene	0.12	vibration/rotation of chain molecules
Polyethylene	0.46-0.50	
Polystyrene	0.13	
Teflon	0.25	

Selected values from Table 19.1, *Callister & Rethwisch 8e.*

Thermal Conductivity Comparison



HfB_2/SiC materials have relatively high thermal conductivity

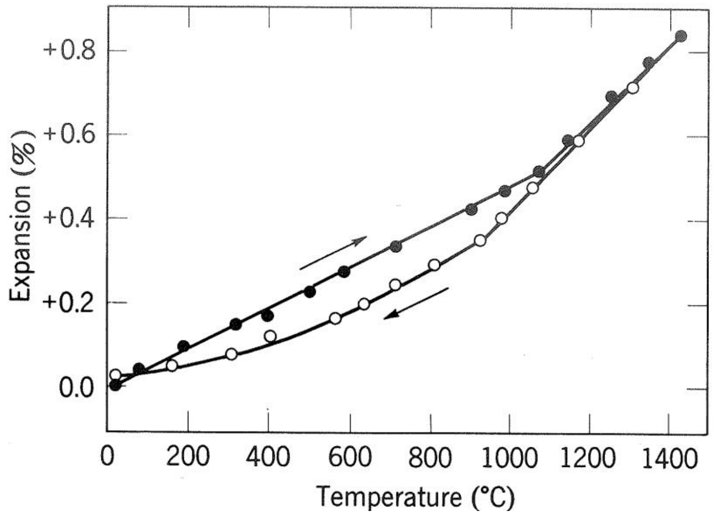
Thermal Stresses

- Occur due to:

-- restrained thermal expansion/contraction

-- temperature gradients that lead to differential dimensional changes

Thermal stress $$\sigma = E\alpha_\ell (T_0 - T_f) = E\alpha_\ell \Delta T$$



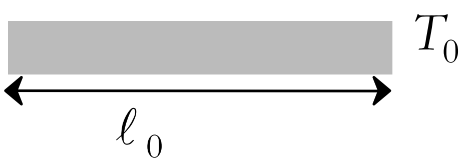
Occurs particularly in ploycrystalline compositions when the expansion coefficient is markedly different in different crystalline directions.

Thermal expansion hysteresis of polycrystalline TiO_2 by presence of microfissures

Example Problem

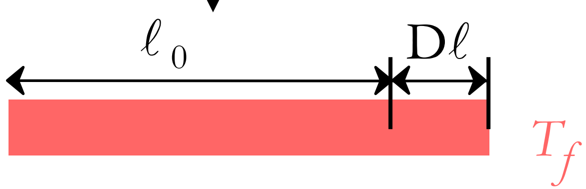
-- A brass rod is stress-free at room temperature (20°C).
-- It is heated up, but prevented from lengthening.
-- At what temperature does the stress reach -172 MPa?

Solution:



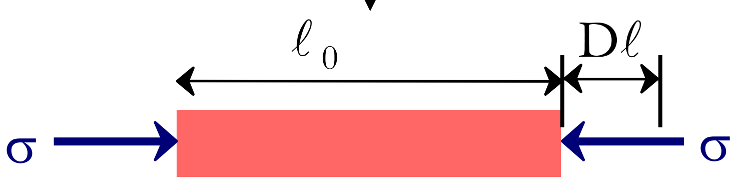
Original conditions

Step 1: Assume unconstrained thermal expansion



$$\frac{\Delta \ell}{\ell_{\text{room}}} = \varepsilon_{\text{thermal}} = \alpha_\ell (T_f - T_0)$$

Step 2: Compress specimen back to original length



$$\varepsilon_{\text{compress}} = \frac{-\Delta \ell}{\ell_{\text{room}}} = -\varepsilon_{\text{thermal}}$$

Example Problem (cont.)

The thermal stress can be directly calculated as

$$\sigma = E(\varepsilon_{compress})$$

Noting that $\varepsilon_{compress} = -\varepsilon_{thermal}$ and substituting gives

$$\sigma = -E(\varepsilon_{thermal}) = -E\alpha_\ell(T_f - T_0) = E\alpha_\ell(T_0 - T_f)$$

Rearranging and solving for T_f gives

20°C

-172 MPa (since in compression)

$$T_f = T_0 - \frac{\sigma}{E\alpha_\ell}$$

Answer: 106°C

100 GPa

20 x 10^{-6}/°C

Glaze Stresses

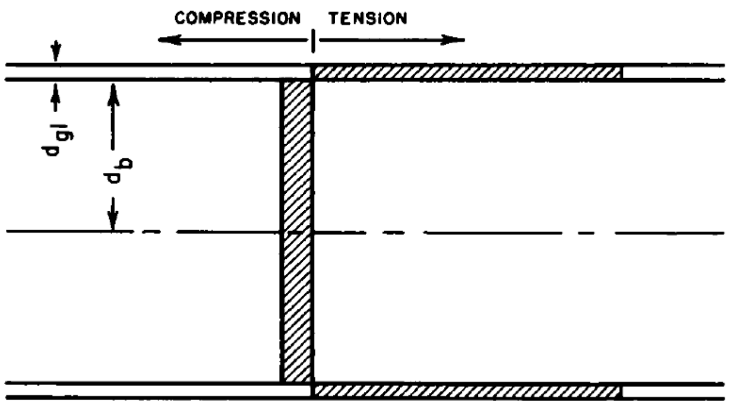


Fig. 1. Stresses in glaze on infinite slab.

For a thin glaze on an infinite slab:

$$\sigma_{gl} = E(T_0 - T')(a_{gl} - a_b)(1 - 3j + 6j^2)$$

$$\sigma_b = E(T_0 - T')(a_b - a_{gl})(j)(1 - 3j + 6j^2)$$

where $j = d_{gl}/d_b$

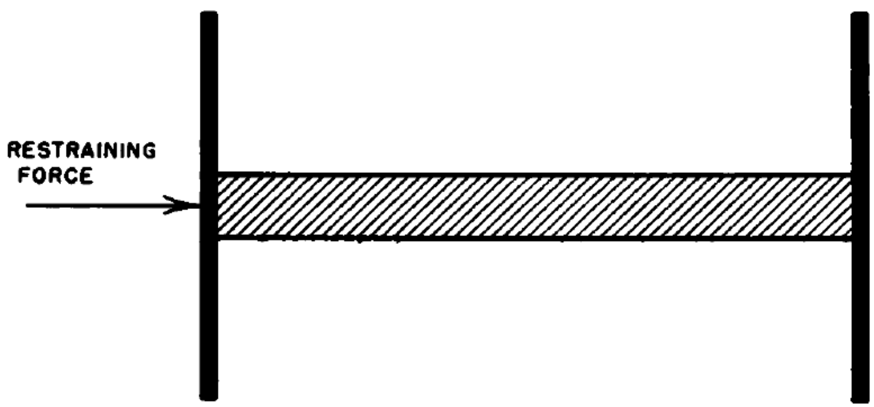


Fig. 2. Restraint of expansion by fixed supports.

If a bar of material is completely restrained from expanding by application of restraining forces

$$\sigma = \frac{Ea\,\Delta T}{1 - \mu}$$

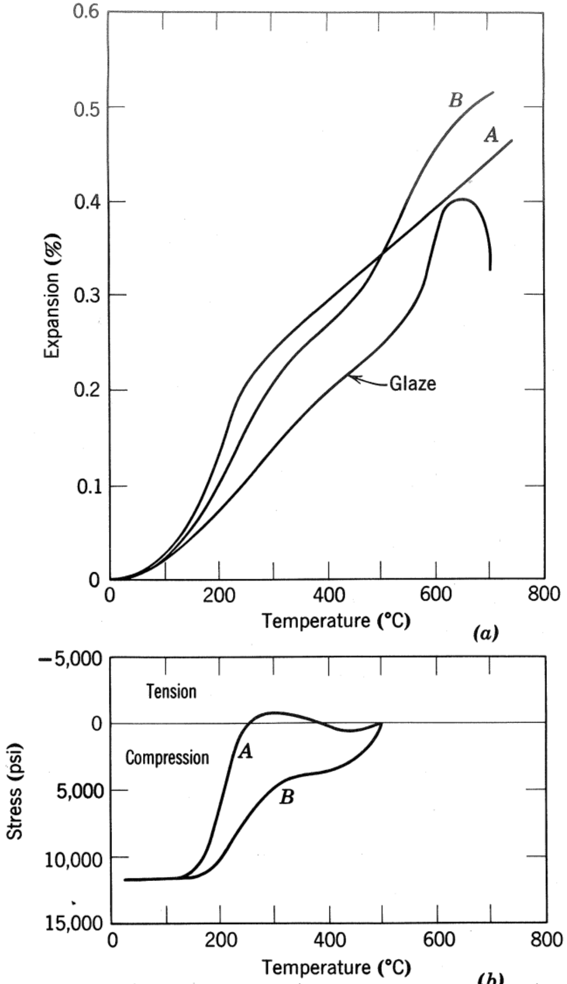
Glaze Stresses

For a thin glaze on a cylindrical body:

$$\sigma_{gl} = \frac{E}{1-\mu}(T_0 - T')(a_{gl} - a_b)\frac{A_b}{A}$$

$$\sigma_b = \frac{E}{1-\mu}(T_0 - T')(a_b - a_{gl})\frac{A_{gl}}{A}$$

Crazed glaze showing tensile cracks

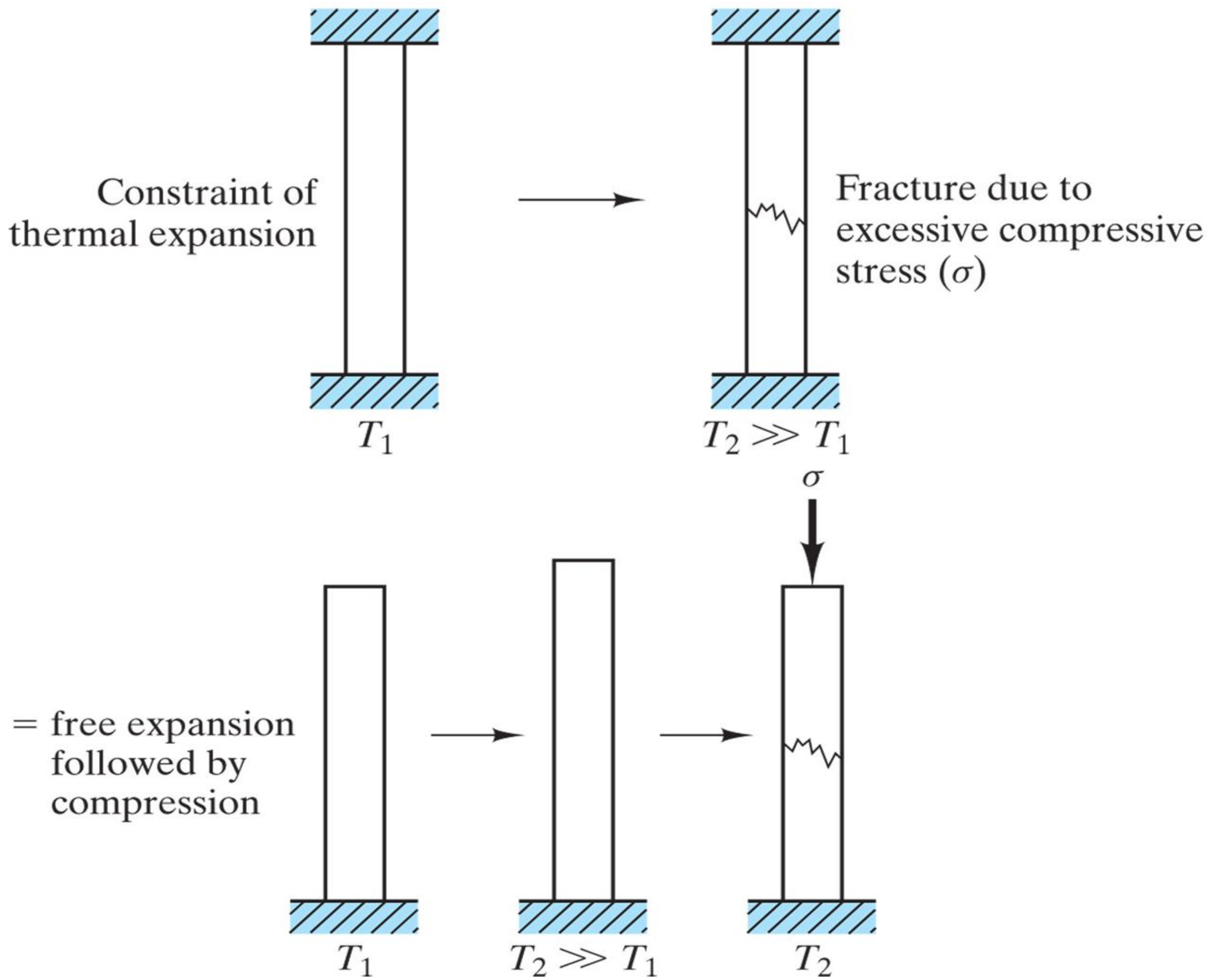


(a) Expansion of glaze and porcelain bodies and (b) stresses developed on cooling.

In order to obtain satisfactory fit between a glaze and body, which is desirable for glaze after cooling to Room Temperature ? Tension or Compression ? Why ? How to achieve ?

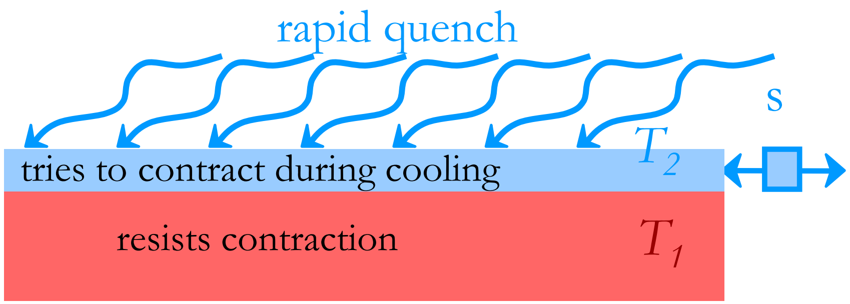
Thermal shock

Thermal shock resulting from constraint of uniform thermal expansion. This process is equivalent to free expansion followed by mechanical compression back to the original length.



Thermal Shock Resistance

- Occurs due to: nonuniform heating/cooling
- Ex: Assume top thin layer is rapidly cooled from T_1 to T_2



Tension develops at surface

$$\sigma = -E\alpha_\ell (T_1 - T_2)$$

Temperature difference that can be produced by cooling:

$$(T_1 - T_2) = \frac{\text{quench rate}}{k}$$

Critical temperature difference for fracture (set s = s$_f$)

$$(T_1 - T_2)_{\text{fracture}} = \frac{\sigma_f}{E\alpha_\ell}$$

set equal

- $$(\text{quench rate})_{\text{for fracture}} = \text{Thermal Shock Resistance } (TSR) \propto \frac{\sigma_f k}{E\alpha_\ell}$$
- Large *TSR* when $\dfrac{\sigma_f k}{E\alpha_\ell}$ is large

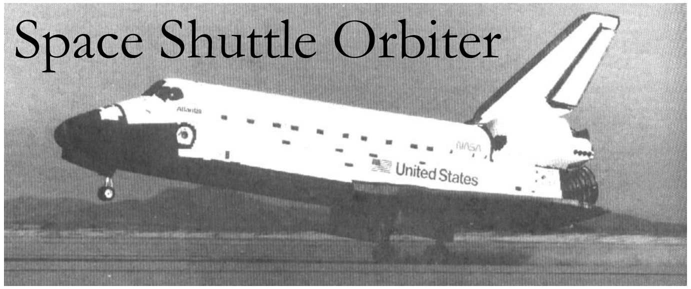
How to prevent failure due to thermal shock ?

Failure due to thermal shock can be prevented by:

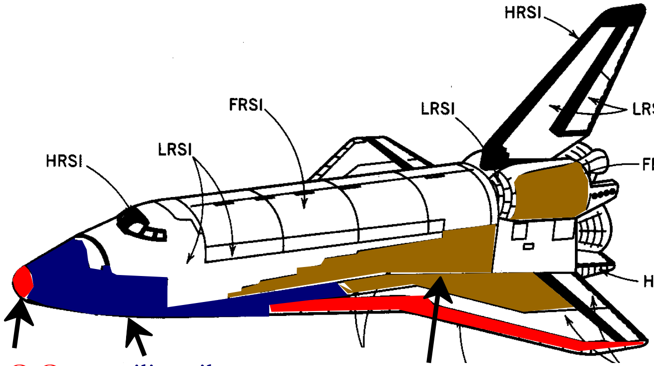
- Reducing the thermal gradient seen by the object, by
  1. changing its temperature more slowly
  2. increasing the material's thermal conductivity
- Reducing the material's coefficient of thermal expansion
- Increasing its strength
- Introducing built-in compressive stress, as for example in tempered glass
- Decreasing its Young's modulus
- Increasing its toughness, by
  1. crack tip blunting, i.e., plasticity or phase transformation
  2. crack deflection

Thermal Protection System

- Application:



Chapter-opening photograph, Chapter 23, *Callister 5e* (courtesy of the National Aeronautics and Space Administration.)



reinf C-C (1650°C)

silica tiles (400-1260°C)

nylon felt, silicon rubber coating (400°C)

Fig. 19.2W, *Callister 6e*. (Fig. 19.2W adapted from L.J. Korb, C.A. Morant, R.M. Calland, and C.S. Thatcher, "The Shuttle Orbiter Thermal Protection System", *Ceramic Bulletin*, No. 11, Nov. 1981, p. 1189.)

- Silica tiles (400-1260°C):

-- large scale application

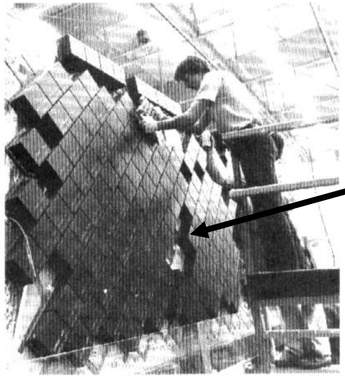
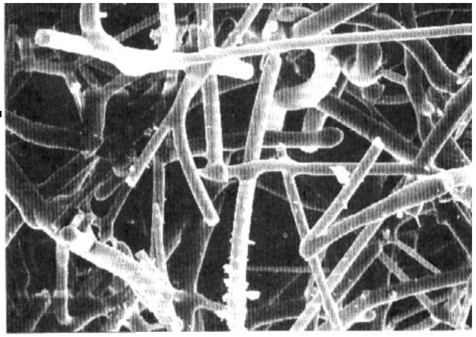


Fig. 19.3W, *Callister 5e*. (Fig. 19.3W courtesy the National Aeronautics and Space Administration.)

-- microstructure:



100 mm

~90% porosity!
Si fibers bonded to one another during heat treatment.

Fig. 19.4W, *Callister 5e*. (Fig. 219.4W courtesy Lockheed Aerospace Ceramics Systems, Sunnyvale, CA.)

Summary

The thermal properties of materials include:

- Heat capacity:
  -- energy required to increase a mole of material by a unit T
  -- energy is stored as atomic vibrations
- Coefficient of thermal expansion:
  -- the size of a material changes with a change in temperature
- Thermal conductivity:
  -- the ability of a material to transport heat
  -- metals have the largest values
- Thermal shock resistance:
  -- the ability of a material to be rapidly cooled and not fracture
  -- is proportional to $\dfrac{\sigma_f k}{E\alpha_\ell}$